

APPENDIX D

Coordinate systems

A right-handed reference frame is defined by an origin and a set of three orthogonal axes in a right-handed set. The origin is often either the center of the Earth or the center-of-mass of the vehicle. In general, however, the origin of the body frame of a vehicle is defined at a selected point, and the center-of-mass varies as fuel is consumed or articulated bodies, such as robot arms, are moved. The point is often chosen during manufacturing as a reference.

D.1. Earth-centered inertial coordinates

The inertial reference frame used is the J2000.0 frame. J2000.0 defines the date January 1, 2000 at 12h Universal Time. This frame has its xy -plane parallel to the mean Earth equator at epoch J2000.0 and its z -axis pointing towards the mean North celestial pole of J2000.0. The x -axis points toward the mean vernal equinox of J2000.0. Greenwich is the meridian that passes through the Royal Observatory in Greenwich in the United Kingdom. This frame is also sometimes called the Earth-Centered Inertial frame (ECI). It is shown in Fig. D.1 with the Earth-Fixed (EF) and local vertical/local horizontal (LVLH) frames.

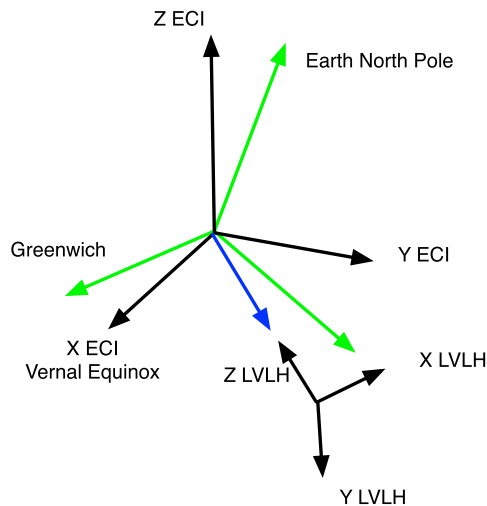


Figure D.1 Coordinate frames.

The black-axis system is the ECI system. The green is EF. The z -axis for the EF system is along the true North pole of the Earth and the x -axis is along the Greenwich meridian. The z -axis for the ECI frame is along the mean North pole of the Earth. The EF system rotates about the z -axis once per day. The deviation between the true North pole and mean North pole is very small and can usually be neglected.

The transformation matrix between the ECI frame and the Earth-Fixed frame is the product of three matrices, G , N , and P

$$M = GNP \quad (\text{D.1})$$

where G is the Earth rotation about the pole, N is the Earth nutation, and P is the Earth precession.

D.2. Local vertical/local horizontal coordinates

The LVLH (local vertical/local horizontal) frame is a rotating frame and is very convenient for Earth-oriented spacecraft. It has the $+z$ -axis pointing at the Earth (nadir), $-y$ along the orbit normal, and $+x$ completing the right-hand set. This frame rotates at orbit rate and is the same as the Earth-centered frame when the vehicle is in an equatorial orbit. “Mean” coordinates for the Earth neglect the Earth’s nutational motion. If Earth nutation is included, the coordinates are known as true.

A few other definitions are in order. Nadir refers to the vector from the center of the vehicle to the center of the Earth. Zenith points away from the Earth. In the Earth-pointing frame nadir is the $+z$ axis and zenith is $-z$. The nadir vector would always be parallel to the vehicle position vector if the Earth were spherically symmetric, see Fig. D.2.

The LVLH matrix (that transforms from ECI to LVLH) is

$$m = \begin{bmatrix} x & y & z \end{bmatrix} \quad (\text{D.2})$$

where

$$y = \frac{v^\times r}{|v^\times r|} \quad (\text{D.3})$$

$$z = -\frac{r}{|r|} \quad (\text{D.4})$$

$$x = \frac{y^\times z}{|y^\times z|} \quad (\text{D.5})$$

where x , y , and z are column vectors that form the matrix. Converting between frames involves computing the rotation matrices that perform the transformation between the frames. Celestial objects are often referred to in the Earth-centered frame.

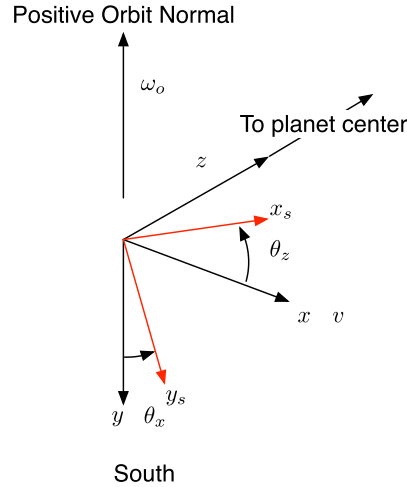


Figure D.2 Local vertical/local horizontal (LVLH) frame.

D.3. Heliocentric coordinates

The heliocentric frame z -axis is normal to the ecliptic plane, that is the plane of the Earth's orbit. The x -axis is along the line of Aries that points through the Earth to the constellation of Aries. The transformation matrix from ecliptic to the Earth equatorial planes is a single rotation about the x -axis.

$$T = (J - 2\,451\,545)/36\,525 \quad (\text{D.6})$$

$$\theta = ((23.43929111 - ((46.815 + (0.00059 - 0.001813T)T)T)/3600) \quad (\text{D.7})$$

where J is the Julian date in days and T is Julian centuries. The transformation matrix is

$$c = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \quad (\text{D.8})$$

D.4. International Space Station coordinates

Many spacecraft need to operate near the International Space Station (ISS).

The ISS frame is defined as:

1. +X Along the velocity vector (Forward);
2. -X Opposite the velocity vector (Aft);
3. +Y Negative Orbit Normal (Starboard);

4. $-Y$ Positive Orbit Normal (Port);
5. $+Z$ Zenith (Deck);
6. $-Z$ Nadir (Overhead).

The coordinates are shown in Fig. D.3.

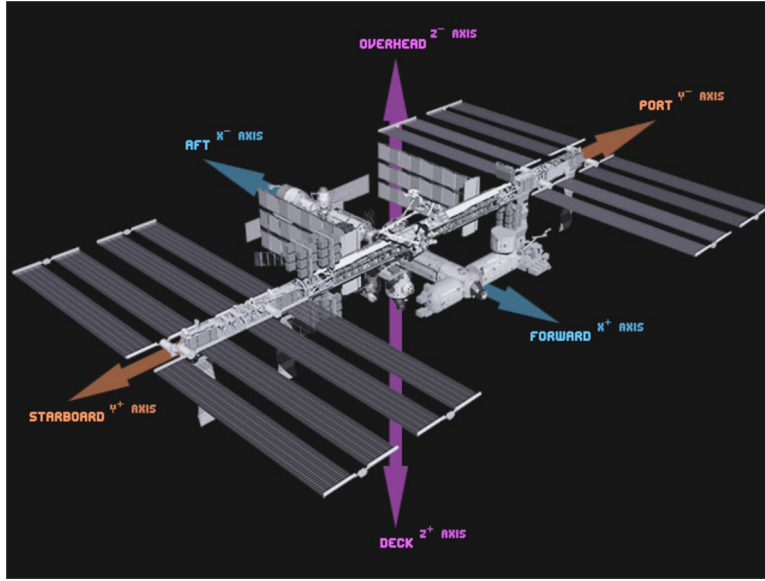


Figure D.3 International Space Station frame. Image Courtesy of NASA.

D.5. Selenographic frame

The Selenographic frame is shown in Fig. D.4. The Moon's prime meridian is the line passing from the lunar North pole to the South pole through the point on the lunar surface directly facing Earth. ϕ'_m is the Selenographic Latitude, which is the acute angle measured normal to the Moon's equator between the equator and a line connecting the geometrical center of the coordinate system with a point on the surface of the Moon. The angle range is between -90 and $+90$ degrees. λ_m is the Selenographic Longitude, which is the angle measured towards the West in the Moon's equatorial plane, from the lunar prime meridian to the object's meridian. The angle range is between 0 and 360 degrees. North is the section above the lunar equator containing Mare Serenitatis. West is measured towards Mare Crisium.

The relationship to the ECI frame is shown in Fig. D.5. i is the inclination between the Moon's equator and the ECI equator. Ω' is the longitude of the mean ascending

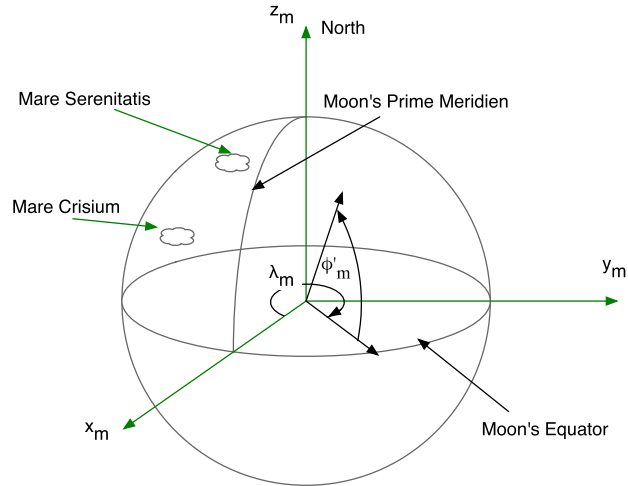


Figure D.4 The selenographic reference frame.

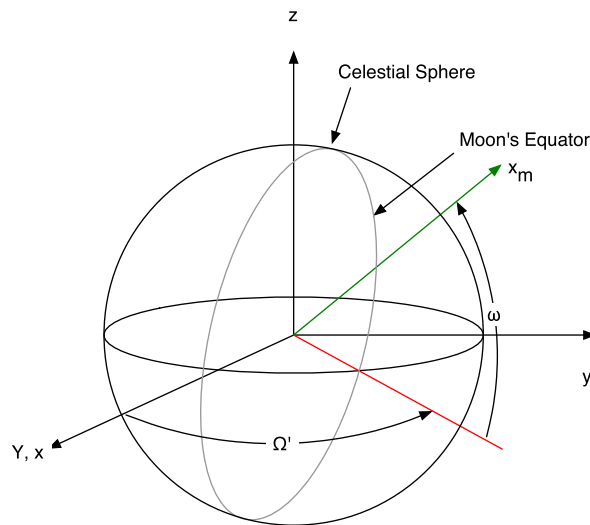


Figure D.5 Selenographic to ECI frame.

node of the lunar orbit referenced to the mean equinox. ω is the angle between the ascending node and the lunar prime meridian.

D.6. Areocentric (Mars) coordinates

The areocentric frame or Mars frame is shown in Fig. [D.6](#).

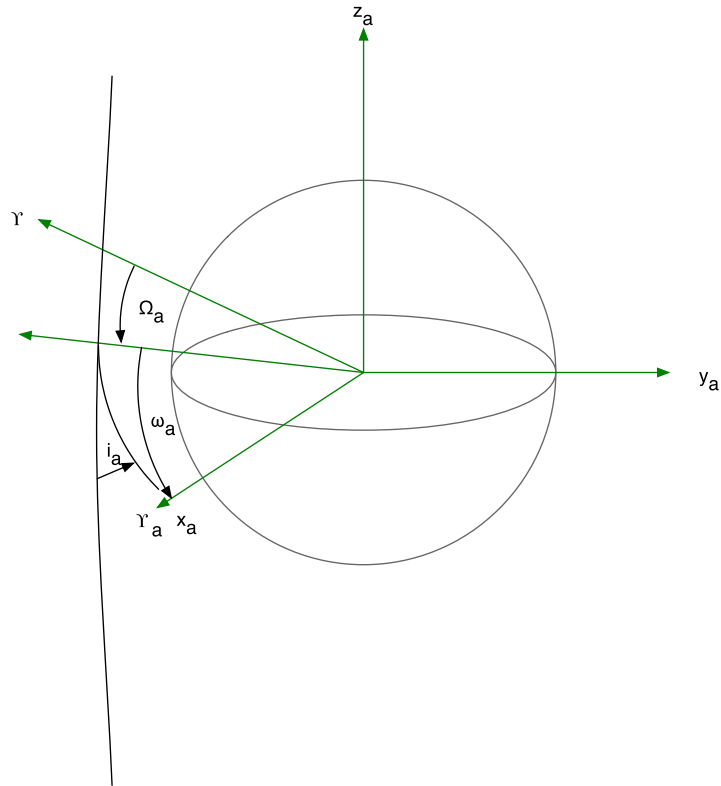


Figure D.6 Areocentric frame.

Ω_a , ω_a , and i_a are the standard Euler rotations of the Mars vernal equinox, Υ_a with respect to the Earth's vernal equinox, Υ .